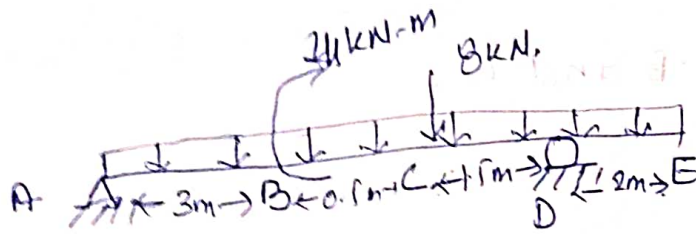
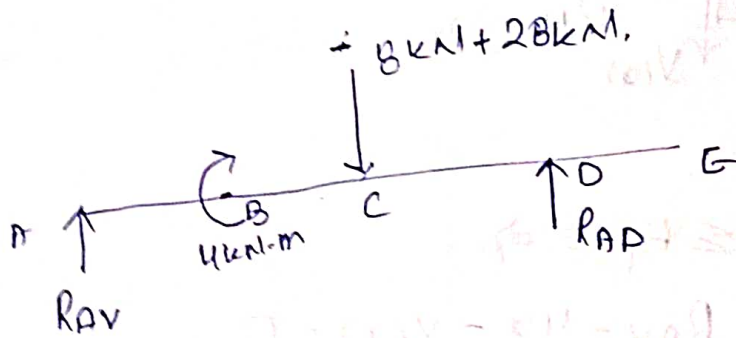


Q.4



①



$$\sum F_y = 0 \Rightarrow R_{AV} - 8 - 4(7) + R_{DV} = 0$$

$$R_{AV} + R_{DV} = 36 \text{ kN.}$$

①

$$\sum M_A = 0 \Rightarrow -4 - (8 + 28)3.5 + R_{DV}(5) = 0$$

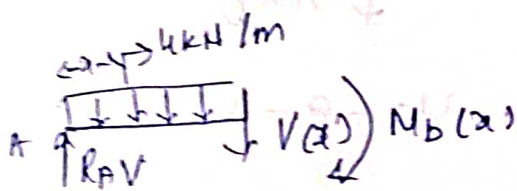
$$R_{DV} = 26 \text{ kN}$$

①

$$R_{AV} = 36 - 26 = 10 \text{ kN.}$$

①

② Cross section b/w A and B -



$$\sum F_y = 0$$

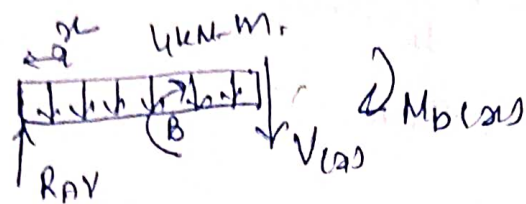
$$R_{AV} - 4x - V(x) = 0$$

$$V(x) = 10 - 4x$$

$$\sum M_x = 0 \Rightarrow M_b(x) + R_{AV}(x) - 4 \times x \times \frac{x}{2} = 0$$

$$M_b(x) = -10x + 2x^2$$

Gross section b/w B and C



$$\sum F_y = 0 \Rightarrow$$

$$R_{AV} - 4x - V(x) = 0$$

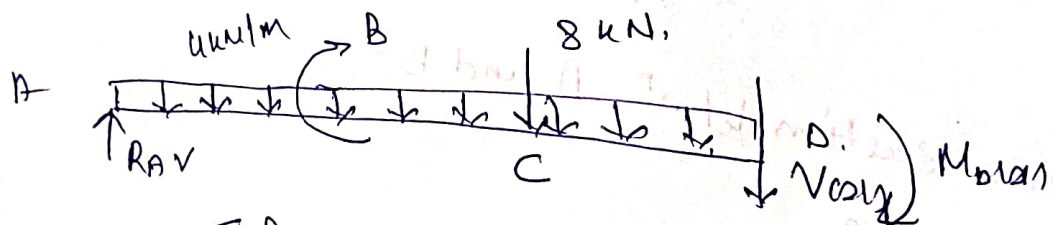
$$V(x) = 10 - 4x$$

$$\sum M_x = 0 \Rightarrow R_{AV}(x) - 4x\left(\frac{x}{2}\right) + M_b(x) + 4 = 0$$

$$\Rightarrow M_b(x) = 2x^2 - 14$$

$$M_b(x) = 2x^2 - 10x - 4$$

Gross section b/w C & D



$$\sum F_y = 0 \Rightarrow R_{AV} - 4x - 8 - V(x) = 0$$

$$\Rightarrow V(x) = 2 - 4x$$

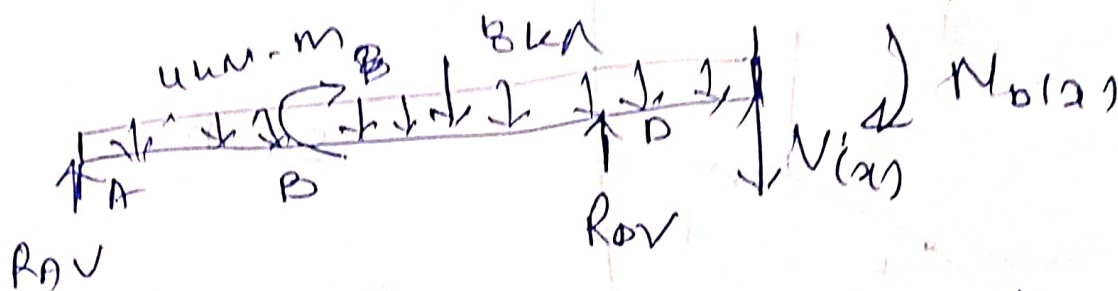
$$\sum M_x = 0 \Rightarrow R_{AV}(x) - 4x \times \frac{x}{2} + 4 - 8(x-1) + M_b(x) = 0$$

$$\Rightarrow M_b(x) = 2x^2 - 2x - 32$$

$$10x - 2x^2 + 4 - 8x + 28 = 0$$

$$\Rightarrow 2x^2 - 2x - 32 = M_b(x)$$

Gross section blwⁿ 0 & E -



$$\sum F_y = 0 \Rightarrow R_{AV} - 4 \times 2 - 8 - V(x) + R_{DV} = 0$$

$$\Rightarrow V(x) = 28 - 4x$$

$$\sum M_A = 0 \Rightarrow R_{AV}(x) - 4x \times \frac{x}{2} + 4 - 8(x - 3) + R_{DV}(x - 5) -$$

$$+ M_b(x) = 0$$

$$10x - 2x^2 + 4 - 8x + 28 \cdot 0$$

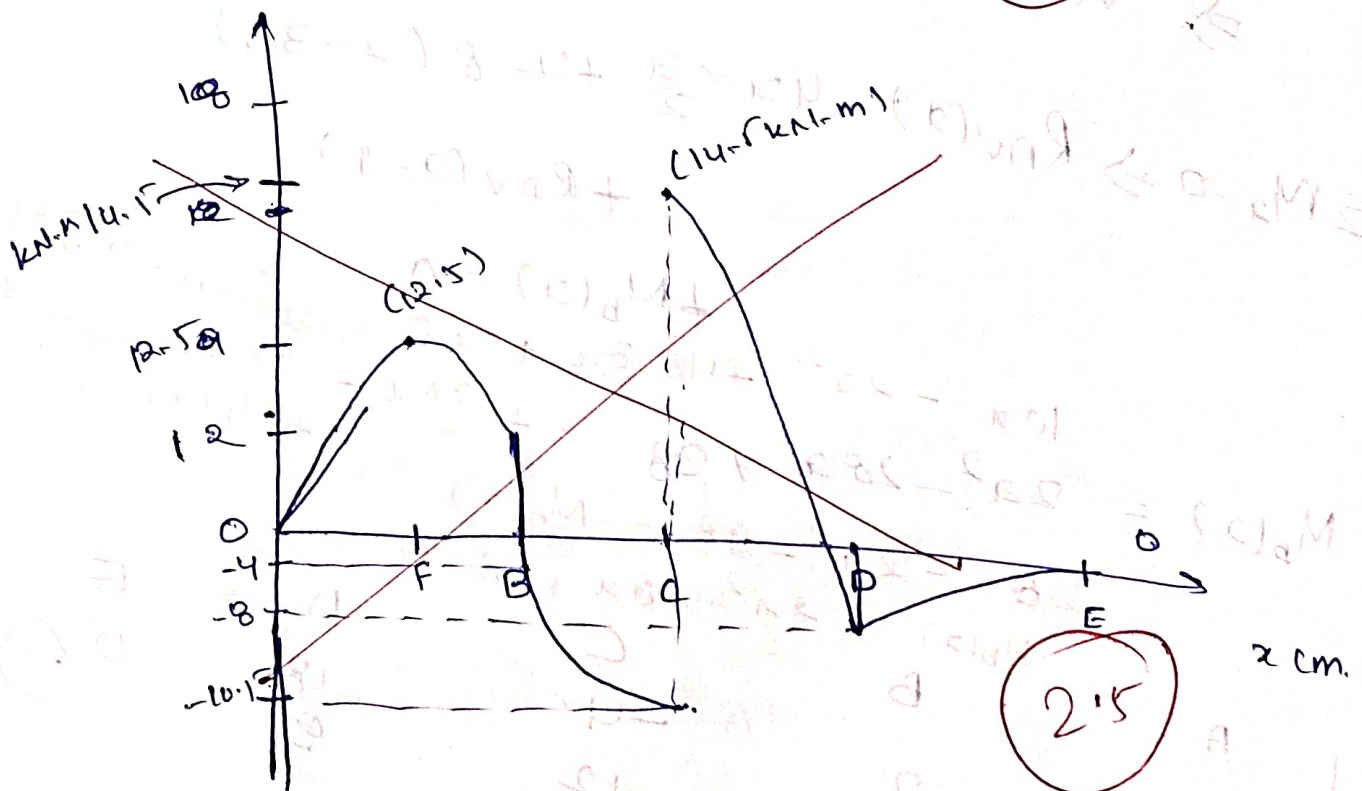
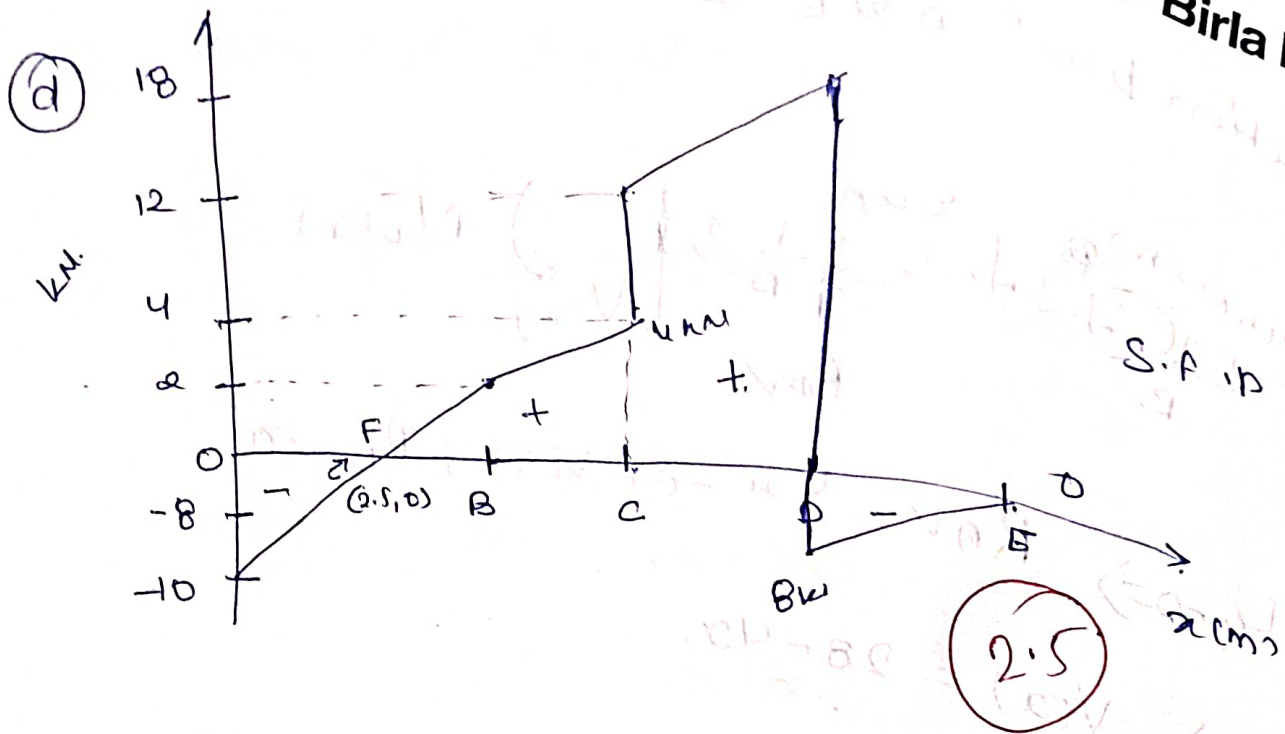
$$M_b(x) = 2x^2 - 28x + 98$$

$$28x - 2x^2 - 98 = -M_b(x)$$

$$M_b(x) : 2x^2 - 28x + 98 = 0$$

Ans.
②

	A	B	C	D	E
$V(x)$ kN	10	-2	-12	-18	0
$M_b(x)$ kN-m	0	+12	+14.5	-8	0

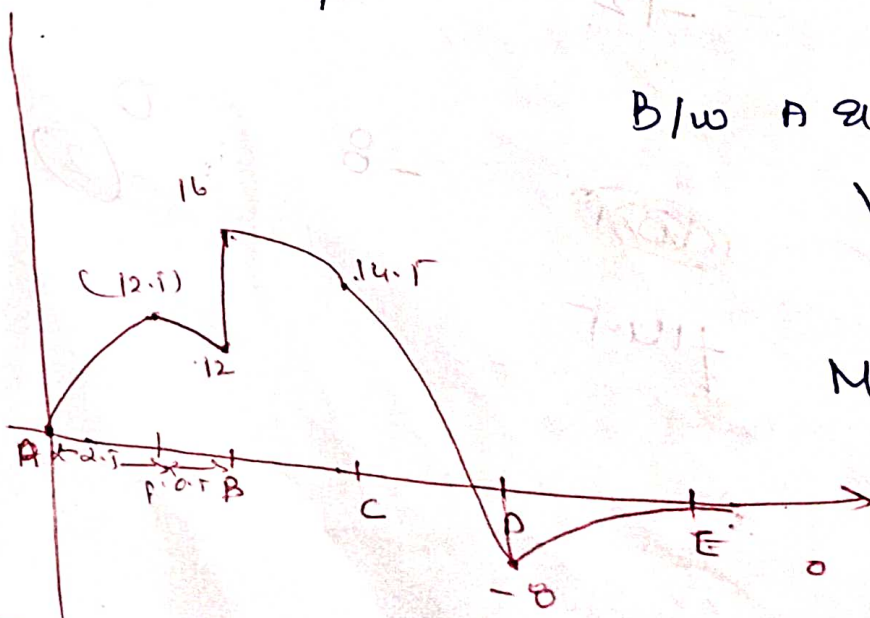


B/w A & B

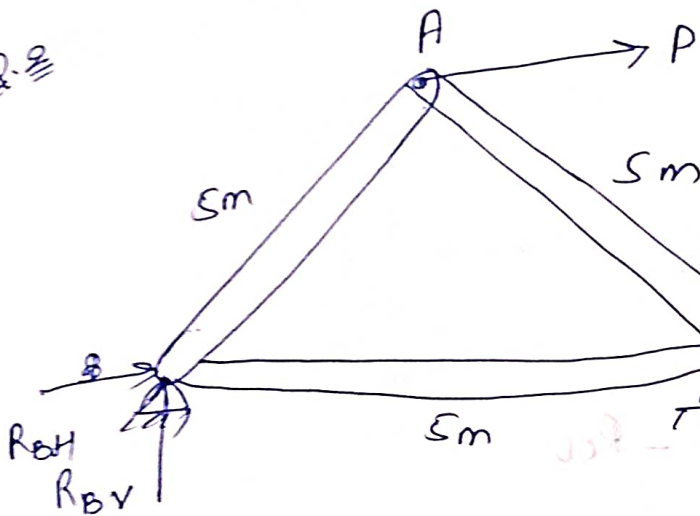
$$V(x) = 0 = 10 - 4x$$

$$x = 10/4 = 2.5$$

$$M_B(2.5) = -12.5 \text{ kN-m}$$



Q. 2



$$E = 205 \times 10^6 \text{ kN/m}^2$$

$$A = 480 \text{ cm}^2$$

$$I_{max} = ?$$

$$R_{CH} = 0$$

$$\delta_c = 0.8 \text{ mm.}$$

Let us assume a horizontal force Q at C
 $CQ = 0 \therefore R_{CH} = 0$

$$\sum F_H = 0$$

$$\Rightarrow R_{BV} = -R_{CV} \quad \text{--- (1)}$$

$$\sum F_V = 0$$

$$\Rightarrow R_{BH} = Q - P \quad \text{--- (2)}$$

$$\sum M_B = 0$$

$$\Rightarrow -P(4.33) + R_{CV}(5) = 0$$

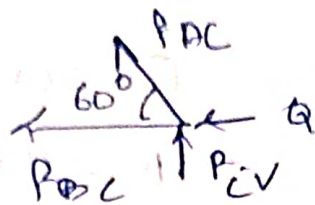
$$\Rightarrow R_{CV} = \frac{4.33P}{5} \quad \text{--- (3)}$$

From 1 & 3 $R_{BV} = -R_{CV} = \frac{4.33P}{5}$

1

1

at C



$$\sum F_y = 0$$

$$\Rightarrow P_{AC} \sin 60^\circ = -P_{CV}$$

$$\Rightarrow P_{AC} = \frac{-4.33}{\sqrt{3}} \times 2 \quad \text{--- (4)}$$



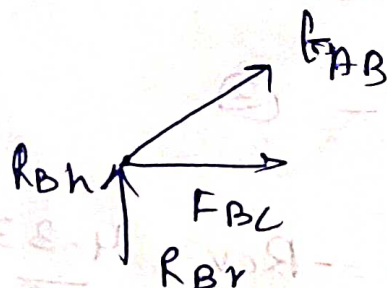
$$\sum F_x = 0 \Rightarrow P_{BC} + Q + P_{AC} \cos 60^\circ = 0$$

$$\Rightarrow P_{BC} = -Q + \frac{4.33}{\sqrt{3}} \times 2 \times P \times \frac{1}{\sqrt{2}}$$

$$= -Q + \frac{4.33}{\sqrt{3}} P \quad \text{--- (5)}$$



at B



$$\sum F_y = 0$$

$$\Rightarrow F_{AB} \sin 60^\circ = -R_{BV}$$

$$\Rightarrow F_{AB} = \frac{-4.33}{\sqrt{3}} \times 2 \quad \text{--- (6)}$$



Now, using Castigliano's theorem

$$\therefore U = \frac{F^2}{2k}, \quad k = \frac{AE}{L}$$

$$U = U_{AB} + U_{BC} + U_{CA}$$

(1)

$$U = \frac{F_{AB}^2 \times L_{AB}}{2 \times A \times E} + \frac{F_{BC}^2 \times L_{BC}}{2 \times A \times E} + \frac{F_{CA}^2 \times L_{CA}}{2 \times A \times E}$$

$$= \frac{L}{2AE} (F_{AB}^2 + F_{BC}^2 + F_{CA}^2)$$

(1)

$$= \frac{L}{2AE} \left[\left(\frac{-4.33 \times 2P}{5\sqrt{3}} \right)^2 + \left(\frac{4.33P}{5\sqrt{3}} - Q \right)^2 + \left(\frac{-4.33 \times 2P}{5\sqrt{3}} \right)^2 \right]$$

(2)

$$\delta_C = \frac{\partial U}{\partial Q} = \frac{L}{2AE} \left[-2 \left[\frac{4.33P}{5\sqrt{3}} - Q \right] \right]$$

(1)

$$\therefore Q = 0$$

$$\delta_C = 0.8 \text{ mm}$$

$$0.8 = \frac{5 \times 10^3 \times 10^6}{2 \times 484 \times 201 \times 10^6 \times 10^3} \left[-2 \left(\frac{4.33P}{5\sqrt{3}} \right) \right]$$

(2)

$$P_{max} = 31.75 \text{ kN}$$

(1)

Ans

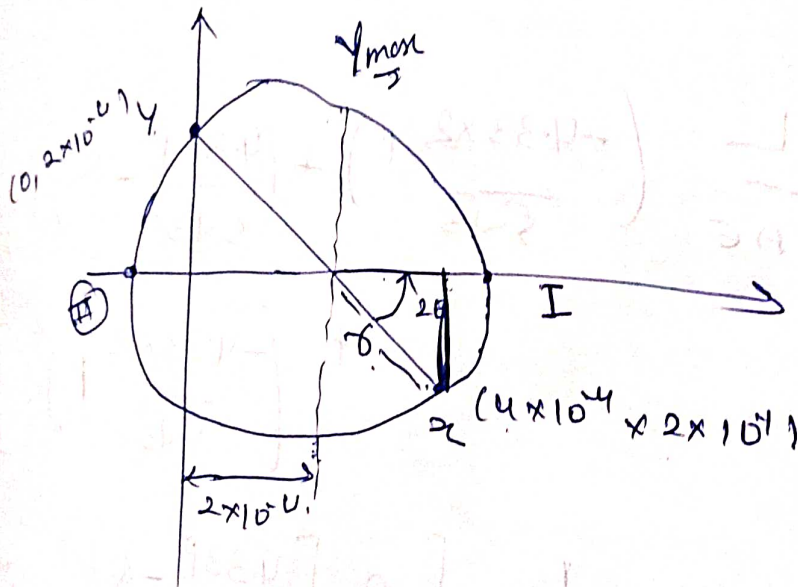
Q.3 Strain = $\frac{\Delta l}{l} = \frac{4 \times 10^{-6}}{10 \times 10^{-3}}$

$\epsilon_{xx} = 4 \times 10^{-4}$

$\Delta \theta = 0.4 \times 10^{-3} \text{ rad}$

$\gamma = 4 \times 10^{-4} \text{ rad}$

Given $\epsilon_{yy} = 0$



$\tan 2\theta = \frac{2 \times 10^{-4}}{2 \times 10^{-4}}$

$2\theta = \pi/4$

$\theta = \pi/8$

$\theta = 22.5^\circ$

antidclockwise
w.r.t. x-y plane

$$\gamma = \sqrt{(2 \times 10^{-4})^2 + (2 \times 10^{-4})^2}$$

$$= \sqrt{4 \times 10^{-8} + 4 \times 10^{-8}}$$

$$= \sqrt{10^{-8} (4+4)}$$

$$= 10^{-4} 2\sqrt{2}$$

$$= 2.82 \times 10^{-4}$$

Principal stress

$$\sigma_1 = 2 \times 10^{-4} + \gamma = 4.82 \times 10^{-4}$$

$$\sigma_2 = 2 \times 10^{-4} - \gamma = -0.82 \times 10^{-4}$$

1 (B)

$$\gamma_{\max} = \frac{\gamma}{2}$$

$$\gamma_{\max} = 2 \times 2.82 \times 10^{-4}$$

$$\gamma_{\max} = 5.64 \times 10^{-4}$$

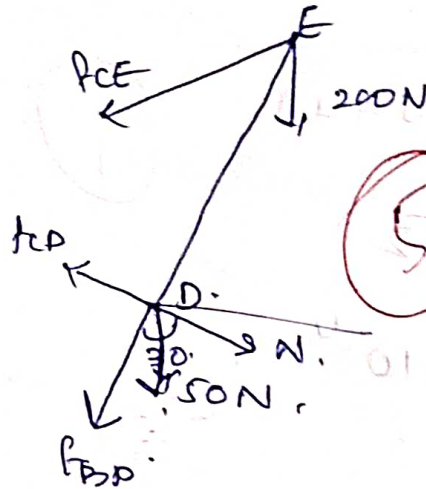
Corresponding Normal stress = 2×10^{-4}

Q1

Method of Joints

$$\sum F_x = 0 \quad 30N = R_x$$

$$\sum F_y = 0 \quad R_y + R_s = 200$$



$$\sum M_E = 0$$

$$F_{CD} \times DE = 200 \sin 30^\circ \times DE = 0$$

$$F_{CD} = 200N$$

Ay

~~10~~